Inflammatory mediators are induced by dietary glycotoxins, a major risk factor for diabetic angiopathy

Diet is a major environmental source of proinflammatory AGEs (heat-generated advanced glycation end products); its impact in humans remains unclear. We explored the effects of two equivalent diets, one regular (high AGE, H-AGE) and the other with 5-fold lower AGE (L-AGE) content on inflammatory mediators of 24 diabetic subjects: 11 in a 2-week crossover and 13 in a 6-week study. After 2 weeks on H-AGE, serum AGEs increased by 64.5% ($P = 0.02$) and on L-AGE decreased by 30% ($P = 0.02$). The mononuclear cell tumor necrosis factor-α/β-actin mRNA ratio was 1.4 ± 0.5 on H-AGE and 0.9 ± 0.5 on L-AGE ($P = 0.05$), whereas serum vascular adhesion molecule-1 was $1{,}108 \pm 429$ and 698 ± 347 ng/ml ($P = 0.01$) on L- and H-AGE, respectively. After 6 weeks, peripheral blood mononuclear cell tumor necrosis factor-α rose by 86.3% ($P = 0.006$) and declined by 20% (P, not significant) on H- or L-AGE diet, respectively; C-reactive protein increased by 35% on H-AGE and decreased by 20% on L-AGE ($P = 0.014$), and vascular adhesion molecule-1 declined by 20% on L-AGE ($P < 0.01$) and increased by 4% on H-AGE. Serum AGEs were increased by 28.2% on H-AGE ($P = 0.06$) and reduced by 40% on L-AGE ($P = 0.02$), whereas AGE low density lipoprotein was increased by 32% on H-AGE and reduced by 33% on L-AGE diet ($P < 0.05$). Thus in diabetes, environmental (dietary) AGEs promote inflammatory mediators, leading to tissue injury. Restriction of dietary AGEs suppresses these effects.